

Theory Is All You Need: AI, Human Cognition, and Causal Reasoning

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Abstract. Scholars argue that artificial intelligence (AI) can generate genuine novelty and new knowledge and, in turn, that AI and computational models of cognition will replace human decision making under uncertainty. We disagree. We argue that AI's data-based prediction is different from human theory-based causal logic and reasoning. We highlight problems with the decades-old analogy between computers and minds as input-output devices, using large language models as an example. Human cognition is better conceptualized as a form of theory-based causal reasoning rather than AI's emphasis on information processing and data-based prediction. AI uses a probability-based approach to knowledge and is largely backward looking and imitative, whereas human cognition is forward-looking and capable of generating genuine novelty. We introduce the idea of data-belief asymmetries to highlight the difference between AI and human cognition, using the example of heavier-than-air flight to illustrate our arguments. Theory-based causal reasoning provides a cognitive mechanism for humans to intervene in the world and to engage in directed experimentation to generate new data. Throughout the article, we discuss the implications of our argument for understanding the origins of novelty, new knowledge, and decision making under uncertainty.



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Introduction

Artificial intelligence (AI) now matches or outperforms humans in any number of games, standardized tests, and cognitive tasks that involve high-level thinking and strategic reasoning. For example, AI engines can readily beat humans in chess, which, for decades, served as a key benchmark of AI capability (Simon 1985a, Bory 2019). AI systems also now perform extremely well in complex board games that involve sophisticated negotiation, complex interaction with others, alliances, deception, and understanding other players' intentions (e.g., Ananthaswamy 2022). Current AI models also outperform more than 90% of humans in various professional qualification exams, such as the bar exam in law and the certified public accountant exam in accounting (Achiam et al. 2023). AI has also made radical strides in medical diagnoses, beating highly trained medical professionals in diagnosing some illnesses (e.g., Zhou et al. 2023). These rapid advances have led some AI scholar to argue that even the most human of traits, such as consciousness,

will soon be replicable by machines (e.g., Goyal and Bengio 2022, Butlin et al. 2023). In all, AI is rapidly devising algorithms that "think humanly," "think rationally," "act humanly," and "act rationally" (Csaszar and Steinberger 2022, pp. 2–3).

Given the astonishing progress of AI, Daniel Kahneman (2018, pp. 609–610, emphasis added) asks (and answers) the logical next question: "Will there be *anything* that is reserved for human beings? Frankly, I don't see any reason to set limits on what AI can do ... And so it's very difficult to imagine that with *sufficient data* there will remain things that only humans can do ... You should replace humans by algorithms whenever possible."

Kahneman is not alone in this assessment. Davenport and Kirby (2016, p. 29) argue that "we already know that analytics and algorithms are better at creating insights from data than most humans" and "this human/machine performance gap will only increase." Many scholars claim that AI is likely to outperform

humans in most—if not all—forms of reasoning and decision making (e.g., Legg and Hutter 2007, Morris et al. 2023, Grace et al. 2024). Some argue that strategic decision making might also be taken over by AI (Csaszar et al. 2024) or even that science itself will be automated by “AI scientists” (e.g., Lu et al. 2024, Manning et al. 2024). One of the pioneers of AI, Geoffrey Hinton, argues that large language models (LLMs) already are sentient and intelligent and “digital intelligence” will inevitably surpass human “biological intelligence”—if it has not already done so (see Hinton 2023; also see Bengio et al. 2023).

Compared with machines, the cognitive and computational limitations of humans are obvious. Humans are biased (Kahneman 2011, Chater et al. 2018). Humans are selective about what data they attend to and sample, and they are susceptible to confirmation bias, motivated reasoning, and hundreds of other cognitive biases (nearly 200 as of last count). In short, humans are boundedly rational—significantly hampered by their ability to compute and process information (Simon 1955), particularly compared with computers (cf. Simon 1990). And the very things that make humans boundedly rational and poor at decision making are seemingly the very things that enable computers to perform well on cognitive tasks. The advantage of computers and AI is that they can handle vast amounts of data and process it quickly and in powerful ways.

In this paper, we offer a contrarian view of AI relative to human cognition, including its implications for strategy, the emergence of novelty, and decision making under uncertainty. AI builds on the idea that cognition—by both machines and humans—is a generalized form of information processing: a type of input–output device. To illustrate cognitive differences between humans and computers, we use the example of large language models versus human language learning. We introduce the notion of data–belief (a)symmetry and the role this, respectively, plays in explaining AI and human cognition, using heavier-than-air flight as an extended example. Human cognition is forward-looking, necessitating data–belief asymmetries, which are manifest in theories, causal reasoning, and experimentation. We argue that human cognition is driven by forward-looking, theory-based causal logic, which is distinct from the emphasis AI and computational models of cognition place on prediction and backward-looking data. Theory-based causal reasoning enables the generation of new and contrarian data, observations, and experimentation. We highlight the implications of these arguments for understanding the origins of novelty, new knowledge, and decision making under uncertainty.¹

AI = Mind: Is Cognition Computation?

Modeling the human mind—thinking, rationality, and cognition—has been the central aspiration and ambition

behind AI from the 1940s to the present (McCulloch and Pitts 1943, Turing 1948/1992; also see Simon 1955, Hinton 1992, McCorduck 2004, Perconti and Plebe 2020). As put by the organizers of the first conference on AI (held at Dartmouth in 1956), their goal was to “proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it” (McCarthy et al. 2007, p. 12). The commonalities between models of AI and human cognition are not just historical, but these linkages have only deepened in the intervening decades (for a review, see Sun 2023; also see Laird et al. 2017). Computation also underlies many other models of cognition, including the concept of mental models (Johnson-Laird 1983), the Bayesian brain, and predictive coding or processing (e.g., Friston and Kiebel 2009; Hohwy 2013, 2020). In fact, cognitive scientist Johnson-Laird (1983, p. 477) goes so far as to argue that “any scientific theory of the mind has to treat it as an automaton.”

AI sees cognition as a general form of computation, specifically in which “human thinking is wholly information-processing activity” (Feigenbaum 1963, p. 249; also see Simon 1980). This logic is also captured by computational neuroscientist David Marr (1982, p. 4), who states that “most of the phenomena that are central to us as human beings—the mysteries of life and evolution, of perception and feeling and thought—are primarily phenomena of information processing” (cf. Hinton 2023). Both mind and machine are a type of generalized input–output device, in which inputs such as stimuli and cues (data) are processed to yield varied types of outputs, including decisions, capabilities, behaviors, and actions (Simon 1980, 1990; McClelland and Rumelhart 1981; Hasson et al. 2020). This general model of information processing has been applied to any number of issues and problems at the nexus of AI and cognition, including perception, learning, memory, expertise, search, and decision making (cf. Russell and Norvig 2022). Furthermore, the idea of human mental activity as computation is pervasive in evolutionary arguments. For example, Cosmides and Tooby (2013, pp. 202–203) focus on the “information-processing architecture of the human brain” and further argue that “the brain is a computer, that is, a physical system that was designed to process information.”

Now, our overall purpose is not to exhaustively review models of AI and cognition, particularly as excellent reviews can be found elsewhere (e.g., Goodfellow et al. 2016, Aggarwal 2018, Russell and Norvig 2022). Rather, we simply want to point out the strong emphasis that past, current, and ongoing research places on the similarities between models of AI and human cognition. To assure the reader that we are not creating a caricature of existing work, we have provided relevant, additional detail about AI-cognition similarities in the appendix.

There, we more exhaustively point out examples of how scholars—from the 1950s to the present—have sought to create an equivalence between AI, machines, and human cognition. In all of this work, cognition and computation (and AI) are seen as deeply connected: the underlying premise of this work is that machines and humans are a form of input–output device, in which the same underlying mechanisms of information processing and learning are at play. The focus on computation and information processing also is the axiomatic basis for the concept of bounded rationality (for a review, see Felin et al. 2017). Bounded rationality focuses on human “computational capacities” and their limits (Simon 1955, p. 99), and this idea has deeply shaped fields such as economics, decision theory, strategy, and the cognitive sciences (e.g., Chater et al. 2018, Kahneman 2003, Puranam et al. 2015, Gigerenzer and Goldstein 2024).

We disagree with the idea that AI and human cognition share significant similarities as forms of computation for reasons to be discussed next. That said, our aim in making this claim is not to take away from the exciting breakthroughs in AI. Rather, we highlight how the analogy between AI and humans quickly breaks down when it comes to understanding the mind and cognition with important derivative consequences for how we think about the emergence of novelty, new knowledge, and decision making under uncertainty. In the next section, we delve into a specific example, namely, language learning by machines versus humans, to enable us to make this point more carefully.

Machine vs. Human Learning: Different Inputs, Different Outputs

While the input–output model of minds and machines—whether we are talking about symbolic or subsymbolic approaches (see the appendix for further detail)—has been a central emphasis of AI and cognitive science, next we highlight some important *differences* between machine learning and human learning. An apt context for highlighting these differences is to focus on language. Language arguably is “the most defining trait of human cognition (language and its relation with thought),” and therefore, it “can be a true ‘window into the mind’” (Chomsky and Gallega 2020, p. 321; also see Pinker 1994).² Language provides an important test and context for understanding human and artificial intelligence. Furthermore, some have already argued that large language models are sentient with a few even arguing that they already closely mirror or exceed human cognition (e.g., Binz and Schulz 2023, Hinton 2023)—an assumption that we challenge.

At the most basic level, to study any system and its behavior, we need to understand its inputs and outputs. Turing (1948/1992) argued that any form of intelligence, whether human or machine, can be studied as an input–output system. In discussing the possibilities of

artificial intelligence—or “intelligent machinery” as he called it—Turing (1950, p. 456, emphasis added) made the analogy to an “untrained infant brain,” saying an infant brain is largely a blank slate, “something like a notebook” with “little mechanism, and lots of blank sheets” (cf. Turing 1948/1992). According to Turing, these blank sheets are (or need to be) filled with inputs via the process of training and education. Through the early course of its life, an infant or child is taught and receives inputs in the form of language and spoken words that it hears and encounters. Education and training represent the inputs that eventually account for human linguistic capacities and outputs. And in the same way, Turing (1948/1992, p. 107) argues, one can think of an “analogous teaching process applied to machines,” where machines learn from their inputs. Turing lists various settings in which a thinking machine might show that it has learned—including games such as chess or poker, cryptography, or mathematics—and he argues that the “learning of languages would be the most impressive, since it is the most human of these activities” (Turing 1948/1992, p. 117). As human and machine learning are often seen as a similar process, we next focus on key differences using language learning as our example. We then highlight the implications of these differences in learning for decision making and knowledge generation both in scientific and economic contexts.

How Machines Learn Language. To illustrate the process of machine learning, next we carefully consider modern LLMs and how they learn. LLMs offer a useful instantiation of machine learning. Learning is essentially generated from scratch—bottom up, directly from the data—through the introduction of vast amounts of training data and the algorithmic processing of the statistical associations and interactions among that data. In the context of an LLM, the training data are composed of enormous amounts of words and text, pulled together from various public sources and the internet. To appreciate just how much data and training these models incorporate, the latest LLMs (as of early 2024) are estimated to include some 13 trillion tokens (a token being the rough equivalent of a word). To put this into context, if a human tried to read this text—say at a speed of 9,000 words/hour (150 words/minute)—it would take more than 164,000 years to read the 13 trillion words of a training data set.

The vast corpus of text used to train an LLM is tokenized to enable natural language processing. This typically involves converting words (or subword units or characters) into numerical sequences or vectors. To illustrate, a sentence such as “The cat sat on the mat” might be tokenized into a sequence such as [10, 123, 56, 21, 90, 78]. Each token is passed through an embedding layer, which converts the token into a dense vector